A Mini Market Intelligence – Netflix Users Database (2024–2025)

INTRODUCTION

This is an capstone Project on Netflix,one of the largest online movie streaming Platforms which aims to focus on **analyze user engagement, content preferences, and subscription trends**, and generate **actionable insights** using:

- **SQL** – for data exploration and transformation
- **Python** – for data wrangling and automation
- **Power BI** – for interactive dashboards

This dataset is an sample model of Netflix users database for 2024 and 2025 worldwide

OBJECTIVE

Through this mini-project, we aim to replicate a **real-world analytics workflow** from data cleaning to visualization

ASK PHASE

This Case study is on Netflix users database analysis which is an Public domain dataset available on Kaggle and GitHub repositories

https://www.kaggle.com/datasets/smayanj/netflix-users-database

Guiding questions

● What is the problem you are trying to solve?

→ Example questions: Defining KPI'S

- Which country has the highest total watch time?
- Who are the top 10 users by total watch duration?
- What is the most watched genre in each region?

PREPARE PHASE

Steps Followed:

1. **Created a New Database**:
    - Named the database **netfix_users** in **MySQL Workbench** using the *Create Schema* option.
2. **Imported the CSV File**:
    - Right-clicked on the **Tables** section inside the **netfix_users** schema.
    - Selected **"Table Data Import Wizard"**.
    - Browsed and selected the dataset CSV file from my system.
    - The table was created and loaded automatically into the schema.
3. **Checked for Null Values**:
    - Ran a query to check for NULLs across key columns like **Age, Country, Subscription_Type**, **watch_time_Hours, Favourite_Genre, and Last_Login**.
    - ✅ **Result:** No NULL values were found

```sql
# check for null rows and columns
SELECT * FROM netflix_users
WHERE
  User_ID IS NULL OR
  Name is NULL OR
  Age is NULL OR
  Country is NULL OR
  Subscription_Type is NULL OR
  Watch_Time_Hours is NULL OR
  Favorite_Genre is NULL OR
  Last_Login is NULL;
```


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

User_ID	Name	Age	Country	Subscription_Type	Watch_Time_Hours	Favorite_Genre	Last_Login

netflix_users 2 ×

Conclusion: The dataset is clean and ready for analysis with no missing or duplicate values. I can now proceed to the next **Phase**

Checking for Duplicate Records

```sql
SELECT
    User_ID,
    COUNT(*) AS DUPLICATE_COUNT
FROM netflix_users
GROUP BY User_ID
HAVING COUNT(*) > 1;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

User_ID	DUPLICATE_COUNT

Conclusion : No Duplicate Records Found

Next i will be moving to process phase to clean and transform the dataset with the help of SQL to present the dataset ready for analytics.

PROCESS PHASE

Checking for Outliers

```sql
SELECT
    MIN(Age) AS Min_Age,
    MAX(Age) AS Max_Age,
    MIN(watch_time_hours) AS Min_Watch,
    MAX(watch_time_hours) AS MAX_watch
FROM netflix_users;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Min_Age	Max_Age	Min_Watch	MAX_watch
13	80	0.12	999.99

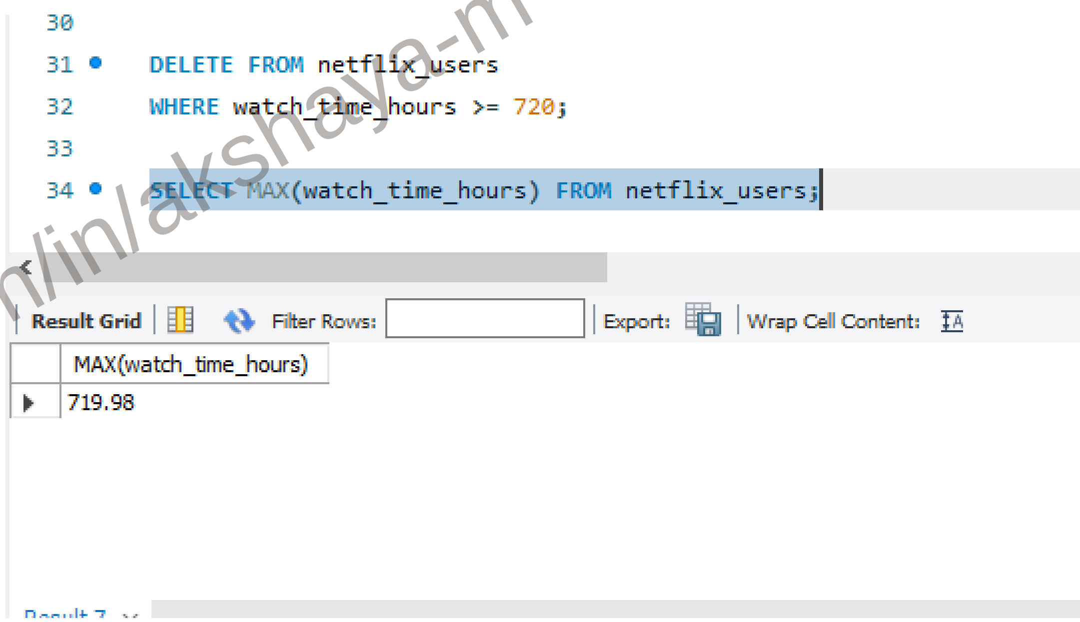
Result 4

- Age
  - Max: **80** → Reasonable range, older viewers.
  - Minimum age Detected at 13
    Netflix officially allows children to watch content under parental supervision via
    Kids' profiles, but creating a standard user account usually requires being **at least 18 years old** (depending on country policies).However this must be verified that these ID's are marked correctly or netflix must ensure data entry on age must be strictly followed

- **Watch_Time_Hours**:
  - Min: **0.12** → Fine (short usage).
  - Max: **999.99** → *Likely an outlier*.

  It is stated in kaggle that the watch_time_hours is for a month and watching 999 Hours in 30 days means approximately watching netflix more than 10 hours a day and thats nearly impossible, so I conclude that the data entered as 999 is an autofill or inaccurate data and these rows would be marked as flagged

VERIFYING MAX VALUES AFTER DELETING EXTREME OUTLIERS



719 watch time hours is Too much extreme for an month for a person to watch Netflix and is unrealistic, so i have decided to delete and update the unrealistic watch time hours in the following steps

Adjusting High Watch Time Values

Action: Updated the watch time to **300 hours** for users exceeding 700.

Adding a New Column – is_outlier

Purpose: To **label** which users are considered outliers based on watch time.

Marking Outliers

Logic:

- Users with watch time **greater than 300** are marked as **outliers** (is_outlier = 1).
- All others are marked **non-outliers** (is_outlier = 0).

```sql
DELETE FROM netflix_users
WHERE watch_time_hours >= 700;

UPDATE netflix_users
SET watch_time_hours = 300
WHERE watch_time_hours > 300;

ALTER TABLE netflix_users
ADD COLUMN is_outlier INT;

UPDATE netflix_users
SET is_outlier = CASE
                   WHEN watch_time_hours > 300 THEN 1
                   ELSE 0
```

CHECKING IF IS_OUTLIER COLUMN IS ADDED

```sql
DESCRIBE netflix_users;
```

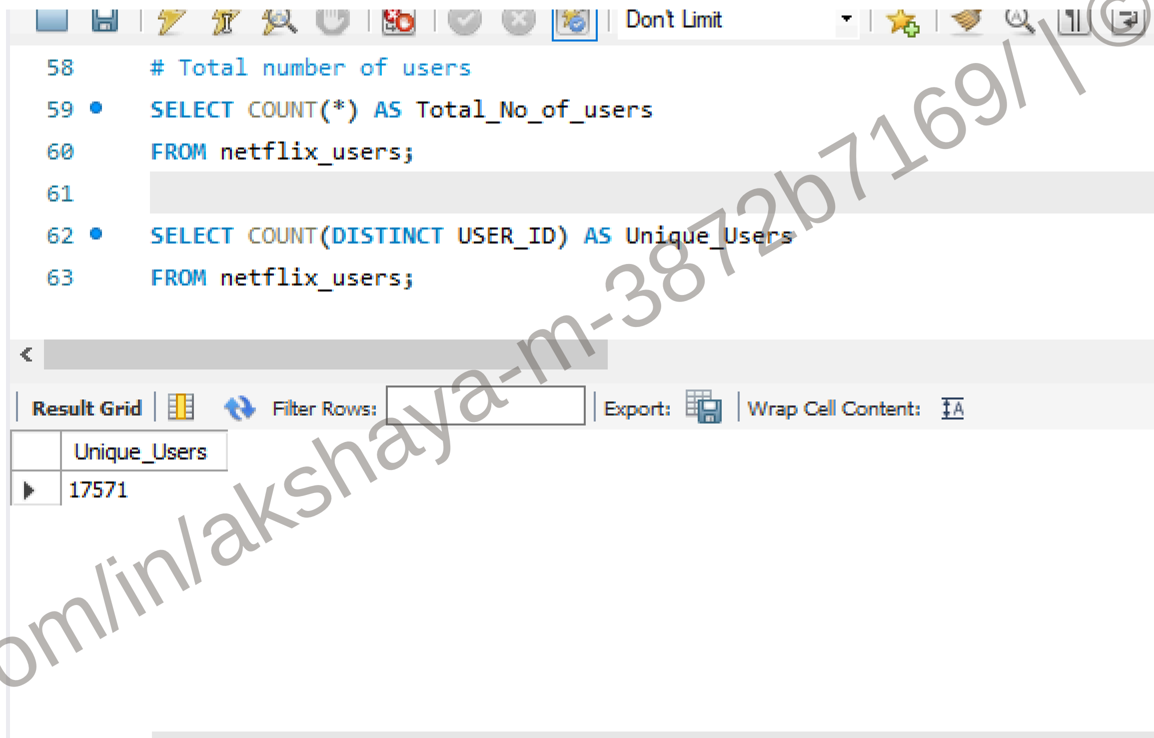
Field	Type	Null	Key	Default	Extra
Country	text	YES		NULL	
Subscription_Type	text	YES		NULL	
Watch_Time_Hours	double	YES		NULL	
Favorite_Genre	text	YES		NULL	
Last_Login	text	YES		NULL	
is_outlier	int	YES		NULL	

Result 18

ANALYZE PHASE

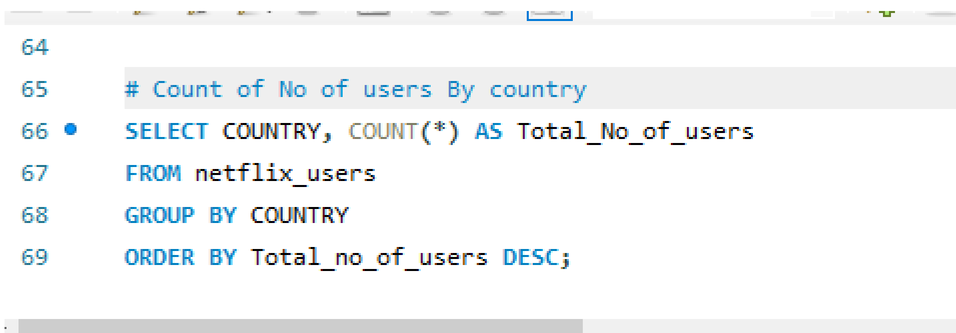
in this phase we are trying to solve some Guiding questions and transform the dataset for analytics with the help of SQL

QUESTION 1: Total Number of Unique users in our sample model dataset of Netflix database(2024 - 2025)



Answer is : The Number of users and the number of unique_users gives the same results 17,571 users and user ID’s

QUESTION 2 : COUNT THE NUMBER OF USERS BY COUNTRY



ANSWER IS :

COUNTRY	TOTAL NO OF USERS
UK	1846
Germany	1800
India	1778
Mexico	1776
USA	1770
Brazil	1751
France	1739
Canada	1728
Japan	1698
Australia	1685

QUESTION 3 : WHAT IS THE AVERAGE WATCH TIME BY SUBSCRIPTION TYPE?

ANSWER IS :

```
# Average watch time hours by subscription type
USE netfix_users;
SELECT Subscription_Type, ROUND(AVG(watch_time_hours), 2) AS Avg_watch_time
FROM netflix_users
GROUP BY Subscription_Type
ORDER BY Avg_watch_time DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Subscription_Type	Avg_watch_time
Basic	238.15
Premium	237.6
Standard	236.49

Logic :

Per day = 24 Hours

Per Month = 30 days or 31 days

so 24 hours x 30 days = 720 Hours in a month

238.15 Hours(in a month) = 238.15/24 = 9.35 days(in a month) which is reasonable

QUESTION 4 : COUNT OF OUTLIERS

ANSWER IS :

```sql
SELECT is_outlier, COUNT(*) AS COUNT
FROM netflix_users
GROUP BY is_outlier;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

is_outlier	COUNT
0	17571

Since I have Updated the maximum outlier as 300 and users with watch time **greater than 300** are marked as **outliers** (is_outlier = 1).

This confirms there is no extreme outlier greater than 300

QUESTION 5 : what is the most popular genre?

THE ANSWER IS :

```sql
# Most popular genre
SELECT Favorite_Genre, COUNT(*) AS genre_count
FROM netflix_users
GROUP BY Favorite_Genre
ORDER BY genre_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Ce

Favorite_Genre	genre_count
Horror	2594
Documentary	2551
Comedy	2516
Action	2515
Romance	2467
Drama	2466
Sci-Fi	2462

QUESTION 6 : who are the users and user_id's who have not logged in the last 30 days?

ANSWER IS:

```sql
# Users who have not logged in the last 30 days
SELECT last_login, COUNT(*) AS not_logged_in
FROM netflix_users
WHERE last_login <= '2025-02-08'
GROUP BY last_login
ORDER BY last_login DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

last_login	not_logged_in
2025-02-08	52
2025-02-07	50
2025-02-06	55
2025-02-05	48
2025-02-04	45
2025-02-03	41
2025-02-02	52
2025-02-01	52

Result 13

QUESTION 7 : CREATING SEGMENTS AS HEAVY VIEWER,MODERATE VIEWER,AND LIGHT VIEWER

THE ANSWER IS:

```sql
SELECt user_ID, Name,
       CASE
          WHEN watch_time_hours >= 200 THEN 'Heavy Viewer'
          WHEN watch_time_hours >= 100 THEN 'Moderate Viwer'
          ELSE 'light viewer'
          END AS viewer_segment
FROM netflix_users;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch r

user_ID	Name	viewer_segment
1	James Martinez	light viewer
2	John Miller	Heavy Viewer
3	Emma Davis	light viewer
4	Emma Miller	Heavy Viewer
6	David Johnson	Heavy Viewer
8	Katie Hernandez	Moderate Viwer
10	Alex Davis	Heavy Viewer
11	Jane Miller	Heavy Viewer

Result 14 ×

so in the analyze phase i have answered some questions and most users in this dataset comes under realistic viewing range,the dataset contains 25,000 records but after removing duplicates the unique user count comes down to 17,571 users

next i am moving to share phase to visualize and communicate insights effectively in Python and Power BI.

SHARE PHASE

Step 1 : importing the dataset in jupyter notebook(python)

I exported the cleaned netflix_users database from SQL to jupyter notebook ,this is an introduction of share phase where where data will be further transformed and visualized using python

The first 5 rows of the code are shown are shown below:

```
[1]: import numpy as np
     import pandas as pd

     df = pd.read_csv(r"D:\netflix_python.csv")
     df.head()
```

[1]:

	User_ID	Name	Age	Country	Subscription_Type	Watch_Time_Hours	Favorite_Genre	Last_Login	is_outlier
0	1	James Martinez	18	France	Premium	80.26	Drama	2024-05-12	0
1	2	John Miller	23	USA	Premium	300.00	Sci-Fi	2025-02-05	0
2	3	Emma Davis	60	UK	Basic	35.89	Comedy	2025-01-24	0
3	4	Emma Miller	44	USA	Premium	261.56	Documentary	2024-03-25	0
4	6	David Johnson	21	USA	Standard	300.00	Romance	2025-02-03	0

Step 2: df.info()

The datatypes and total records and memory usage of the dataset is shown below:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17571 entries, 0 to 17570
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   User_ID             17571 non-null  int64  
 1   Name                17571 non-null  object 
 2   Age                 17571 non-null  int64  
 3   Country             17571 non-null  object 
 4   Subscription_Type   17571 non-null  object 
 5   Watch_Time_Hours    17571 non-null  float64
 6   Favorite_Genre      17571 non-null  object 
 7   Last_Login          17571 non-null  object 
 8   is_outlier          17571 non-null  int64  
dtypes: float64(1), int64(3), object(5)
memory usage: 1.2+ MB
```


Step 3 : **df.describe**()

it describes the mean, standard deviation etc. of the columns as listed below:

```
df.describe()
```

	User_ID	Age	Watch_Time_Hours	is_outlier
count	17571.000000	17571.000000	17571.000000	17571.0
mean	12502.247282	46.383871	237.415887	0.0
std	7221.595621	19.572444	92.659130	0.0
min	1.000000	13.000000	0.120000	0.0
25%	6217.000000	29.000000	180.740000	0.0
50%	12544.000000	46.000000	300.000000	0.0
75%	18762.500000	63.000000	300.000000	0.0
max	25000.000000	80.000000	300.000000	0.0

Step 4 : df.isnull.sum()

```
df.isnull().sum()
```

```
User_ID              0
Name                 0
Age                  0
Country              0
Subscription_Type    0
Watch_Time_Hours     0
Favorite_Genre       0
Last_Login           0
is_outlier           0
dtype: int64
```

Step 5 : Finding out the most common subscription type

```
subscription_type = df['Subscription_Type'].value_counts()
print(subscription_type)
```

```
Subscription_Type
Premium     5910
Basic       5856
Standard    5805
Name: count, dtype: int64
```

Step 6 : Finding out the average watch time by country

```
Avg_watch_time_by_country = df.groupby('Country')['Watch_Time_Hours'].mean().sort_values(ascending=False)
print(Avg_watch_time_by_country)
```

```
Country
France       243.004491
Canada       241.516453
Australia    240.155970
Brazil       239.004175
USA          237.314537
Mexico       237.304150
UK           236.922470
Germany      235.923139
India        232.398645
Japan        230.757344
Name: Watch_Time_Hours, dtype: float64
```


Step 7 : Finding out the Total users by genre

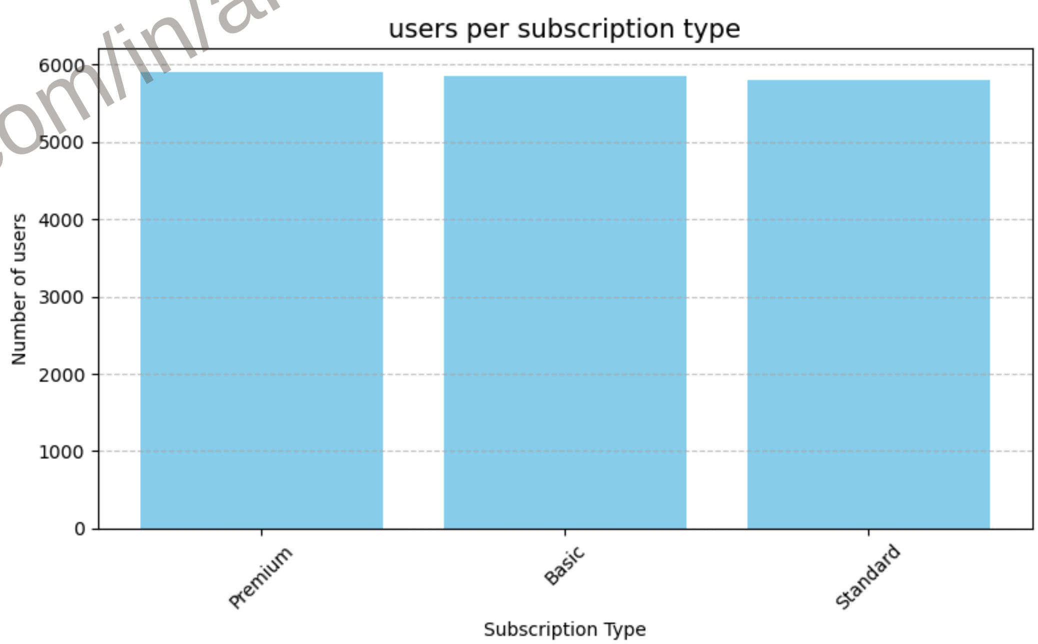
```
favourite_genre = df['Favorite_Genre'].value_counts()
print(favourite_genre)
```

```
Favorite_Genre
Horror         2594
Documentary    2551
Comedy         2516
Action         2515
Romance        2467
Drama          2466
Sci-Fi         2462
Name: count, dtype: int64
```

in the next steps i will be visualize the above questions in share phase using matplotlib with the help of jupyter notebook

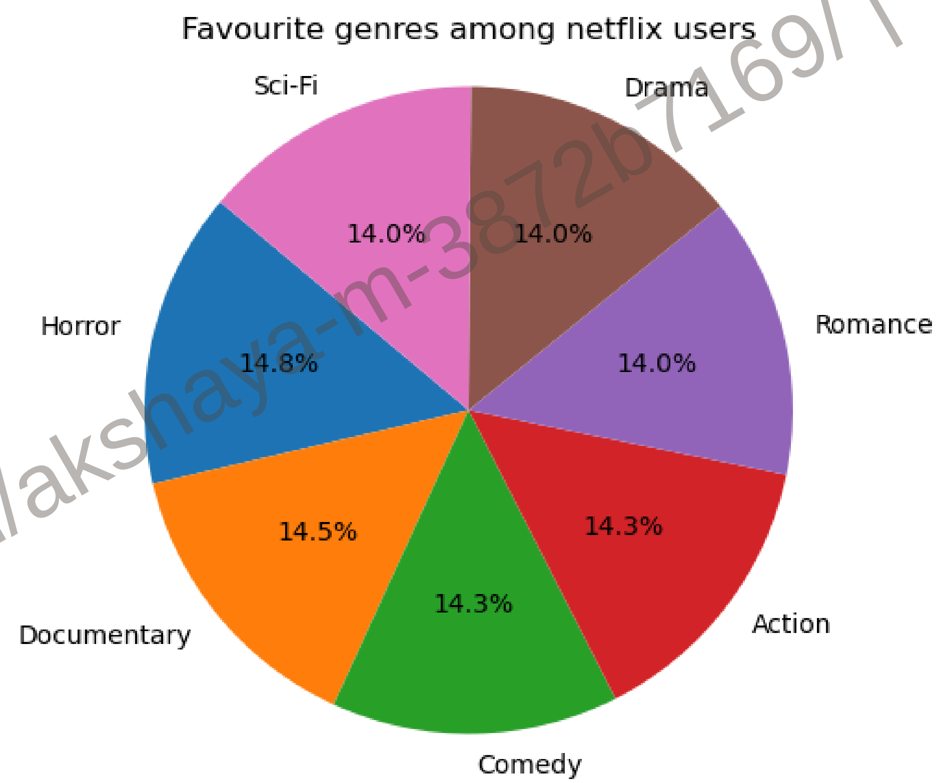
1.Count of users per subscription type

```python
import matplotlib.pyplot as plt

# Count of users per subscription type
subscription_counts = df['Subscription_Type'].value_counts()

# Plotting the Bar Charts
plt.figure(figsize=(8, 5))
plt.bar(subscription_counts.index, subscription_counts.values, color='skyblue')
plt.title('users per subscription type', fontsize=14)
plt.xlabel('Subscription Type')
plt.ylabel('Number of users')
plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



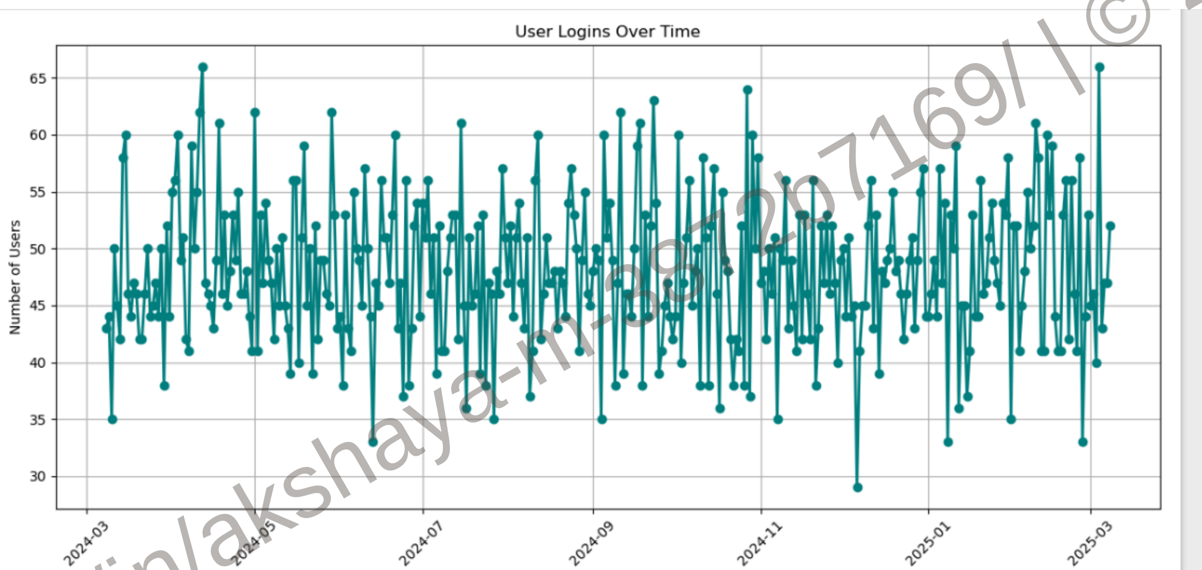
2. Distribution of favourite genre

```python
# Count of each favorite genre
genre_counts = df['Favorite_Genre'].value_counts()


plt.figure(figsize=(8, 5))
plt.pie(genre_counts, labels=genre_counts.index, autopct='%1.1f%%', startangle=140)
plt.title('Favourite genres among netflix users')
plt.axis('equal')
plt.show()
```



3. **Users Login Over Time**

```
# Step 1: Convert 'Last_Login' to datetime
df['Last_Login'] = pd.to_datetime(df['Last_Login'])

# Step 2: Count users not logged in by date
login_counts = df['Last_Login'].value_counts().sort_index()

# Step 3: Plotting
plt.figure(figsize=(12, 6))
plt.plot(login_counts.index, login_counts.values, marker='o', color='teal', linewidth=2)
plt.title('User Logins Over Time')
plt.xlabel('Date')
plt.ylabel('Number of Users')
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
```



Key findings in share phase

(by watch time): Users from USA, Canada, and UK display higher average watch time, indicating strong content engagement.

Drama, Action, and Comedy are the most favored genres, reflecting viewer preferences for emotionally rich and exciting content.

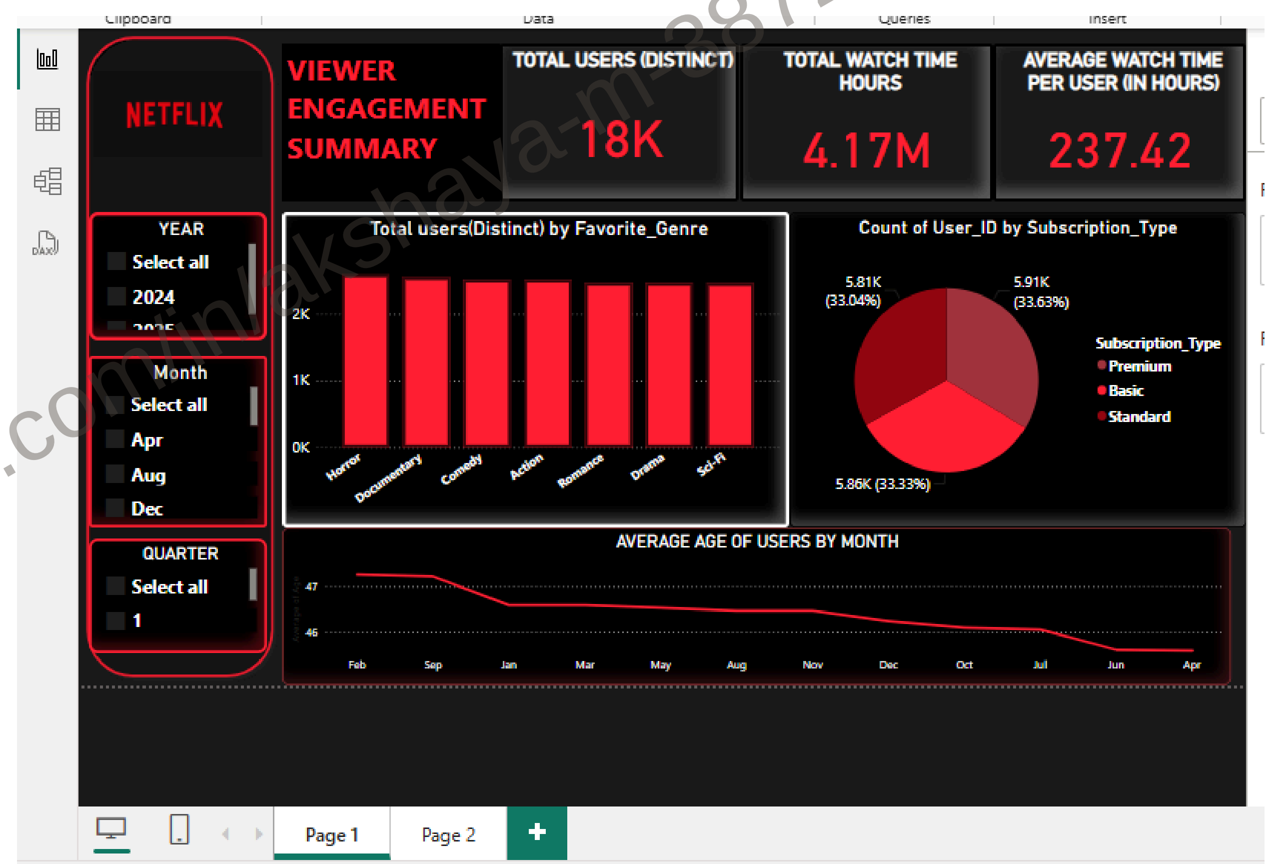
High watch-time countries should be targeted with
localized content and offers.

Consider
personalized recommendations driven by genre preferences.

In the next phase I will be using **netflix_cleaned** data to create visualizations with the help of **POWER BI**

ACT PHASE

In the Act Phase we Transition Code into strategic action That is Visualizations which will be useful for the stakeholders to Pinpoint the the insights uncovered during the analysis of user behavior, viewing patterns, and country-specific trends, this phase highlights key metrics and visualizations that can drive data-informed decisions for Netflix.



This page 1 Refers to some of the KPI's Such as

- 👥 **Total Users (Distinct)** –

  Reflects audience size and engagement base.

- ⏱️ **Total Watch Time (in Hours)** –

  Measures overall user consumption behavior.

- **Average Watch Time Per User (in Hours)** –

  Indicates how engaged each user is.

- **Favorite Genre (Bar Chart)** –

Shows content preferences at scale.

- **Subscription Type Distribution (Pie Chart) –**

  Breaks down user revenue segments.

- **Average Age of Users by Month (Line Chart) –**

  Highlights shifts in demographics over time.

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Conclusion

UK and **Germany** are the leading contributors in terms of viewer engagement.

Next comes **japan** and **Australia** where Netflix can concentrate more on promotions.

Over to the Next Countries Netflix has to focus bringing more local content and monitoring Consistency and retention rates